

CONCEPT NOTE

Project Title: PathPilot AI : A Multi-Agent Career Intelligence System for Personalized Learning

1. Background / Introduction

Education plays a vital role in improving employability and economic growth. However, despite the availability of abundant online learning resources, many students struggle to identify the right learning path, relevant skills, and practical projects required for career growth. This lack of personalized mentorship leads to fragmented learning and inefficient preparation for industry roles.

This project aligns with Sustainable Development Goal 4, which focuses on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all.

2. Problem Statement

Students often face difficulty in selecting appropriate learning resources, identifying skill gaps, and building industry-relevant portfolios. Most learners rely on random tutorials or generic courses without a structured roadmap. As a result, they may spend significant time learning irrelevant content, remain underprepared for technical interviews, and struggle to secure internships or jobs.

3. Objective

The objective of PathPilot AI is to develop an Agentic AI-based mentorship platform that provides personalized career guidance and structured learning roadmaps to students. The system aims to help learners identify missing skills, recommend suitable resources, suggest impactful projects, and improve interview readiness.

4. Proposed Solution / Methodology

PathPilot AI uses a multi-agent architecture in which specialized AI agents collaboratively assist the user.

The system consists of:

- Supervisor Agent : analyzes user goals and decides workflow execution
- Skill Gap Agent : identifies missing technical skills
- Resource Recommendation Agent : suggests relevant learning materials
- Project Recommendation Agent : recommends resume-worthy projects
- Interview Preparation Agent : generates interview questions and mock tasks

The Supervisor Agent autonomously coordinates these agents to generate a personalized action plan, making the system a true agentic AI application.

5. Expected Outcome / Impact

PathPilot AI is expected to improve access to quality education by providing scalable, affordable, and personalized mentorship. The system can help students learn more efficiently, build stronger portfolios, and enhance employability by aligning their learning journey with industry requirements.

6. Technology Stack

- Frontend: Streamlit
- Backend: Python
- Generative AI: Google Gemini API
- Agent Orchestration: Multi-Agent Workflow
- Version Control: GitHub

7. Future Scope

Future enhancements may include resume parsing, GitHub profile analysis, real-time job market integration, long-term memory, and mobile deployment.